NATHISHWAR C

Generative AI Engineer | AI Applications Engineer | LLM & RAG Developer | FastAPI | Python

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Portfolio | Linked in | Github

SUMMARY

AI & Machine Learning Engineering student specializing in **Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, and scalable AI applications**. Experienced in designing production-ready AI systems using **Python, FastAPI, LangChain, Hugging Face, PostgreSQL, Redis, Docker, and FAISS**. Skilled in building intelligent document search, AI tutoring systems, semantic retrieval pipelines, and automation workflows. Passionate about solving real-world business problems through modern AI engineering, scalable backend architecture, and production-quality software development.

SKILLS SUMMARY

Programming: Python, SQL, JavaScript , TypeScript
Artificial Intelligence : Machine Learning, Deep Learning, Neural Networks, Feature Engineering, Model Evaluation, Data Preprocessing ,Model Deployment, Scikit-Learn, PyTorch
Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG),AI Agents,Prompt Engineering,Prompt Chaining,Context Engineering ,Semantic Search,Embeddings,Vector Search,Document Intelligence,Knowledge Retrieval,LLM Evaluation,Hallucination Reduction,OpenAI API,Google Gemini API,Hugging Face Transformers,LangChain,LangGraph
Backend: FastAPI, REST APIs , Python , Authentication , JWT , Microservices
Databases: PostgreSQL, Redis, FAISS , Vector DB
DevOps: Docker, Linux, CI/CD , Github actions , Vercel , Render , AWS (Basic)
Tools: Git, GitHub, Postman, VS Code

PROJECTS

KnowGrow AI — Adaptive Learning Intelligence Platform
• Architected a production-ready AI learning platform integrating **LLMs, Retrieval-Augmented Generation (RAG), AI tutoring, and semantic search** for personalized education.
• Designed a modular FastAPI backend supporting authentication, AI tutoring, course management, quizzes, revision tracking, and progress analytics.
• Built an adaptive recommendation engine using confidence-based spaced repetition algorithms and personalized learning intelligence.
• Integrated vector search using FAISS embeddings to retrieve relevant educational content with low-latency semantic retrieval.
• Engineered AI-powered content generation workflows capable of producing quizzes, explanations, summaries, and learning paths.
• Containerized backend services using Docker with PostgreSQL and Redis to support scalable deployment.
• Designed RESTful APIs following API-first architecture for future web and mobile integration.

AI Knowledge Assistant (RAG-based)
• Developed an enterprise Retrieval-Augmented Generation (RAG) application enabling intelligent document-based question answering.
• Built automated document ingestion pipelines supporting parsing, chunking, embedding generation, indexing, and semantic retrieval.
• Implemented vector similarity search using FAISS and Hugging Face embedding models to improve contextual document retrieval.
• Connected LLM inference with retrieval pipelines to reduce hallucinations and improve factual accuracy.
• Designed scalable FastAPI APIs supporting multi-document querying, conversational memory, and AI-assisted knowledge retrieval.
• Structured the application following modular service architecture for future enterprise deployment.

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WORK EXPERIENCE

AI/ML Intern — Google Developer Student Club (GDSC)
Oct 2024 – Dec 2024
• Engineered end-to-end Machine Learning pipelines using Scikit-Learn for predictive analytics and classification tasks.
• Performed feature engineering, preprocessing, model training, evaluation, and hyperparameter optimization.
• Built reusable Python modules for data preprocessing and inference workflows.
• Collaborated in Agile development using Git-based version control and code reviews.
• Applied software engineering best practices to improve code quality and maintainability.

EDUCATION

B.Tech Artificial Intelligence and DataScience — **9.0 CGPA**
Dhanalakshmi Srinivasan Engineering College — Aug 2023 - May 2027
Marutham matric.Hr.Sec.School — **89.8%**
Bio Mathematics — Jun 2021 - May 2023

CERTIFICATES

- Machine Learning - Stanford & Deeplearning.ai
- Prompt Engineering — Google
- AI Essentials - Google
- Problem Solving — HackerRank